Solution

FORCE-02

Class 09 - Science

Section A

1. State True or False:

(i) **(b)** False
Explanation: False. The second law of motion states that the rate of change of momentum of an object is proportional to the applied unbalanced force in the direction of the force.

(ii) **(b)** False
Explanation: False. Heavy objects have more inertia than lighter objects.

(iii) **(a)** True
Explanation: True

(iv) **(a)** True
Explanation: True

(v) **(a)** True
Explanation: True

(vi) **(a)** True
Explanation: True

(vii) **(a)** True
Explanation: True

(viii) **(a)** True
Explanation: True

(ix) **(a)** True
Explanation: True

(x) **(a)** True
Explanation: True

(xi) **(b)** False
Explanation: False. The momentum of a body at rest is zero.

(xii) **(a)** True
Explanation: True

(xiii) **(a)** True
Explanation: True

(xiv) **(b)** False
Explanation: False. The SI unit of force is kg ms^{-2}.

(xv) **(b)** False
Explanation: False. The SI unit of acceleration is ms^{-2}.

(xvi) **(a)** True
Explanation: True

Section B

2. Fill in the blanks:

(i) 1. Unbalanced

(ii) 1. Safety belts

(iii) 1. Balanced

(iv) 1. kg

(v) 1. Momentum

(vi) 1. Forward

(vii) 1. Friction

(viii)1. Balanced

(ix) 1. Unbalanced

(x) 1. First

(xi) 1. Vector

(xii) 1. N

(xiii)1. Opposing

(xiv)1. Inertia

(xv) 1. Inertia

(xvi)1. Vector

(xvii)1. Third

(xviii)1. Inertia

(xix)1. Zero

Section C

3. The correct order of match is given as (a) – (ii), (b) – (i), (c) - (iv), (d) – (iii).
4. The correct order of match is given as (a) – (ii), (b) – (i), (c) - (iv), (d) – (iii).
5. The correct order of match is given as (a) – (ii), (b) – (i), (c) - (iv), (d) – (iii).
6. The correct order of match is given as (a) – (ii), (b) – (i), (c) - (iv), (d) – (iii).
7. The correct order of match is given as (a) – (ii), (b) – (iii), (c) – (iv), (d) – (i).
8. The correct order of match is given as (a) – (iv), (b) – (iii), (c) - (ii), (d) – (i).
9. The correct order of match is given as (a) – (iii), (b) – (iv), (c) – (i), (d) – (ii).
10. The correct order of match is given as (a) – (iii), (b) – (iv), (c) - (ii), (d) – (i).

Section D

11.

(b) 1-B, 2-D, 3-A, 4-C

Explanation:

- Inertia depends on the mass of object.
- Friction is a necessary evil because neither movement of bodies not holding anybody would have been possible without friction.
- Momentum can be given as the product of mass and velocity.
- Force can be defined as the rate of change of momentum.

12. **(a)** 1-D, 2-A, 3-C, 4-B

Explanation:

- Any body is accelerated due to force acting on it respecting the first law of motion.
- Mass is the quantitative measure of inertia.
- Powder reduces the friction on the carom board.
- Tyres of vehicles are made rough to increase the friction between road and vehicle.

13.

(c) (a) - (iii), (b) - (ii),(c) - (iv), (d) - (i)

Explanation:

- The force the horse exerts on the cart is of equal size and opposite direction to the force the cart exerts on the horse, by Newton's third law.
- When a bus or train starts to move suddenly, the passengers sitting gets a backward jerk. This is due to inertia called inertia of rest.
- The soles of shoes are worn out due to the effect of friction.
- Recoiling of guns is an example of the law of conservation of momentum.

14.

(b) 1-C, 2-B, 3-D, 4-A

Explanation:

- Frictional force is a contact force that opposes the motion of a body.
- The force which is just enough to bring about change in state and tend a body to motion is called limiting force of friction.
- The force of friction which is just sufficient to make a body slide over any surface is called sliding friction.
- rolling friction acts upon when a body rolls over any surface.

15. **(a)** (i)-(d), (ii)-(a), (iii)-(c), (iv)-(b)
Explanation:
    i. Force can be defined as the product of mass and acceleration i.e. F = m $\times$ a.
    ii. Momentum(p) defined as the product of mass and velocity i.e. p = m $\times$ v.
    iii. Change in momentum can be given as- Δp = $\frac{p_2 - p_1}{t}$
    iv. The law of conservation of momentum, momentum before the collision is equal to momentum after the collision.

16.
(b) 1-D, 2-A, 3-C, 4-B
Explanation:
- **Newton's First Law of Motion**: Any object remains in the state of rest or in uniform motion along a straight line until it is compelled to change the state by applying the external force.
- **Newton's Second Law of Motion**: The rate of change of momentum is directly proportional to the force applied in the direction of the force.
- **Newton's Third Law of Motion**: There is an equal and opposite reaction for every action.
- **Galileo Galilei:** Galileo first of all said that objects move with a constant speed when no forces act on them. This means if an object is moving on a frictionless path and no other force is acting upon it, the object would be moving forever. That is there is no unbalanced force working on the object.

17.
(c) 1-A, 2-C, 3-B, 4-D
Explanation: Units of-
- force- Newton
- momentum- kg m/s
- acceleration is caused due to unbalanced force acting on the body.
- rocket works on the principle of Newton's third law of motion

18.
(b) 1-B, 2-D, 3-A, 4-C
Explanation:
- Cart is pulled by horse due to Newton's third law of motion.
- Backward jerk is felt on an account of the law of inertia.
- Soles of shoes are worn out due to the force of friction while walking.
- Recoiling of guns after a fire occurs with respect to the law of conservation of momentum.

19.
(d) (a) - (iv), (b) - (i),(c) - (iii), (d) - (ii)
Explanation:
    i. Friction is the force that makes it difficult for one object to slide along the surface of another.
    ii. The limiting friction is the frictional force just sufficient to move the object.
    iii. Sliding friction refers to the resistance created by two objects sliding against each other.
    iv. Rolling friction refers to the resistance created by an object rolling across a surface.

20.
(d) (a) - (iv), (b) - (i),(c) - (iii), (d) - (ii)
Explanation:

- Inertia is the resistance of any physical object to any change in its state of motion.
- Friction is the necessary evil.
- Momentum = Mass × Velocity
- Force - Rate of change of momentum (Newton's Second law).

21.

(b) 1-B, 2-D, 3-A, 4-C

Explanation:

- An unbalanced force cause change in the state of motion i.e. acceleration in a body.
- Inertia has an approach to remain conserved.
- Grease reduces friction which may lead to wear and tear of machine parts.
- Force can be defined as the rate of change of momentum.

22. **(a)** 1-C, 2-B, 3-D, 4-A

Explanation:

- According to Newton's third law in every interaction, there is a pair of forces acting on the two different interacting objects.
- Newton's second law states that the acceleration of an object as produced by a net force is directly proportional to the magnitude of the net force, in the same direction as the net force, and inversely proportional to the mass of the object.
- Momentum is a vector quantity. For a particle with mass, the momentum equals mass times velocity, and velocity is a vector quantity while mass is a scalar quantity. A scalar multiplied by a vector is a vector.
- The resultant of a balanced force has no acceleration.

23.

(b) (a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)

Explanation:

i. Forces that cause a change in the motion or state of an object are unbalanced forces.
ii. Inertia is a tendency to do nothing or to remain unchanged.
iii. One of the methods of reducing friction is by applying grease.
iv. Force is the push or pulls that is applied to an object to change its momentum.

24. **(a)** (a) - (iv), (b) - (i),(c) - (iii), (d) - (ii)

Explanation:

- Unbalanced forces cause objects to accelerate with an acceleration that is directly proportional to the net forc**e** and inversely proportional to the mass.
- Mass is the quantitative or numerical measure of a body's inertia.
- Power reduces the friction between the carom board and the striker so that strikers can slide on the board smoothly.
- Tyres of vehicles are made rough to increase friction.

Section E

25.

(c) Both statement A and B are true

Explanation: The rocket contains fuel as well as oxygen to burn its fuel as combustion takes place only in the presence of oxygen. A jet plane is filled with hydrogen only and it takes oxygen from the atmosphere to burn its fuel. So, both statements are true.

26.

(b) statement B is true

Explanation: A passenger falls backward when a bus suddenly starts moving in the forward direction due to the inertia of rest. A gun recoils backward with a small speed than the bullet moving forward due to the law of conservation of momentum.

27.

(c) Both statements A and B are true.

Explanation: Gravitational force is a very weak force as it depends upon the mass of the body and distance between two objects. The electrostatic force is a non-contact force as it acts even if there is no contact between the objects.

Section F

28.

(d) (A)

Explanation: The spring force is the force exerted by a compressed or stretched spring upon any object that is attached to it. An object that compresses or stretches a spring is always acted upon by a force that restores the object to its rest or equilibrium position. So, the spring force does not always push a body.

29.

(b) A, C and D

Explanation: Frictional force always produces retardation while gravitational force also produces acceleration. The frictional force will act when two surfaces are in contact while Gravitational force acts when the object is at some height.

30.

(b) act on the same bodies

Explanation: Newton’s third law of motion explains action and reaction. Action and reaction act simultaneously, are equal in magnitude. They act in different directions and acts on different bodies.

31.

(c) A, B and D

Explanation: The object moving along a straight path with an accelerated motion has changing velocity, speed and mass.

32.

(c) (A), (B) and (C) are correct

Explanation: Since the body is accelerating it means that a force acting on the body is changing the velocity of the body. Since there is no change in direction the speed is also changing. And mass always remains constant.

33. **(a)** A and C

Explanation: Action and reaction are equal and opposite and act on two different bodies simultaneously.

34.

(c) (A), (C) and (D) are correct

Explanation: Statement (B) is wrong as when you drop a ball from a height gravity provides acceleration.
Also, the frictional force is a retarding force while gravitational force may be retarding or accelerating. The frictional force is a contact force whereas gravitational force acts from distance like from height.

35. **(a)** A and C

Explanation: Newton's third law states that for every force there is a reaction force that is equal in size but opposite in direction.

36.

(b) their sum on/by a body is zero

Explanation: According to Newton's 3rd law of motion, action and reaction are of equal magnitude and directed in the opposite direction.

e.g if action force F_{ab} and reaction force F_{ba}

then, $F_{ab} + F_{ba} = 0$

Section G

37. i. (c) Newton’s third law of motion
ii. (d) (I) and (III)
iii. (d) 500 N
iv. (c) 0.25
v. (a) move away from the shore

38. i. (a) balanced forces act on the balloon
ii. (b) change in the shape of the body
iii. (a) (I), (II), and (III)
iv. (b) 0 N
v. (d) A force is always acting on it

39. i. (d) All of these
ii. (b) A force can accelerate a lighter vehicle more easily than a heavier vehicle that is moving

iii. (a) 24N

iv. (d)

v. (a) Balanced forces act on the balloon

40. i. (b) (I) and (III)

ii. (d) all of the above

iii. (c) Newton's first law of motion

iv. (a) inertia of motion

v. (b) 15000 kg m/s